

Algorithm Analysis and Time complexity

How to Analyze an Algorithm

{ Algorithm Swap(a, b)

temp = a; ——— 1

a = b; ——— 1

b = temp; ——— 1

}

$$f(n) = 3$$

x = 5*a + 6*b ——— 1

space

a — 1

b — 1

temp — 1

$$S(n) = 3$$

Frequency Count Method

Algorithm Sum(A, n)

```
S = 0; _____ 1
for (i = 0; (i < n); i++) _____ n+1
{
    S = S + A[i]; _____ n
}
return S; _____ 1
```

$f(n) = 2n + 3$
 $O(n)$

A

8	3	9	7	2
0	1	2	3	4

n = 5

Algo
{

Space

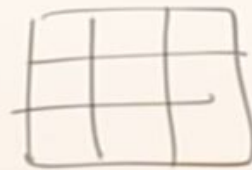
A — n
n — 1
S — 1
i — 1

$S(n) = n + 3$ }
 $O(n)$

Frequency Count Method

Algorithm Add(A, B, n)

$n \times n$
 3×3



for($i=0; i < n; i++$) ————— $n+1$

for($j=0; j < n; j++$) ————— $n \times (n+1)$

$C[i][j] = A[i][j] + B[i][j];$ ————— $n \times n$

$$f(n) = 2n^2 + 2n + 1$$

$$O(n^2)$$

Space

$$A - n^2$$

$$B - n^2$$

$$C - n^2$$

$$n - 1$$

$$i - 1$$

$$j - 1$$

$$S(n) = 3n^2 + 3$$

$$O(n^2)$$

Frequency Count Method

Algorithm Multiply(A, B, n)
{

Time — $O(n^3)$
Space — $O(n^2)$

Space

A — n^2

B — n^2

C — n^2

n — 1

i — 1

j — 1

k — 1

$S(n) = 3n^2 + 4$

$O(n^2)$

n+1 — for(i=0; i<n; i++)
{

(n+1) n — for(j=0; j<n; j++)
{

n x n — c[i, j] = 0;

(n+1) x n x n — for(k=0; k<n; k++)
{

n x n x n — c[i, j] = c[i, j] + A[i, k] * B[k, j];
}

f(n) = $2n^3 + 3n^2 + 2n + 1$
 $O(n^3)$

for ($i=0$; $i < n$; $i++$)

3

stmt; _____ n

3

_____ ၂

 $O(n)$

for ($i=1$; $i < n$; $i=i+2$)

3

$$Stmt; \quad \frac{n}{2}$$

3

$$\frac{n}{2}$$
 $O(n)$

```

for(i=0; i<n; i++) _____ n+1
{
    for(j=0; j<n; j++) _____ n x (n+1)
    {
        stmt; _____ n x n
    }
}

```

$O(n^2)$

```

for(i=0; i<n; i++)
{
    for(j=0; j<i; j++)
    {
        stmt; _____
    }
}

```

$1 + 2 + 3 + \dots + n = \frac{n(n+1)}{2}$
 $f(n) = \frac{n^2 + 1}{2}$
 $O(n^2)$

i	j	no of time
0	0 x	0
1	0 ✓ 1 x	1
2	0 1 2 x	2
3	0 1 2 3	3
...	...	...
n	...	n

$P=0;$

for($i=1; P \leq n; i++$)

{

$P = P + i;$

}

Assume $P > n$

$$\therefore P = \frac{k(k+1)}{2}$$

$$\frac{k(k+1)}{2} > n$$

$$k^2 > n$$

$$k > \sqrt{n}$$

i	P
1	$0+1=1$

2	$1+2=3$
---	---------

3	$1+2+3$
---	---------

4	$1+2+3+4$
---	-----------

⋮

k	$1+2+3+4+\dots+k$
-----	-------------------

$$O(\sqrt{n})$$

⑤

for($i=1; i < n; i=i*2$)
{

stmt;

}

Assume $i \geq n$

$$\therefore i = 2^k$$

$$\therefore 2^k \geq n$$

$$2^k = n$$

$$k = \log_2 n$$

$$\frac{i}{1}$$

$$1 \times 2 = 2$$

$$2 \times 2 = 2^2$$

$$2^2 \times 2 = 2^3$$

⋮

⋮

$$2^k$$

$$O(\log_2 n)$$

⑤

```
for(i=1; i<n; i=i*2)
{
    stmt;
}
```

$$i = 1 \times 2 \times 2 \times 2 \dots = n$$

$$2^k = n$$

$$k = \log_2 n$$

```
for(i=1; i<=n; i++)
{
    stmt;
}
```

$$i = \underbrace{1 + 1 + 1 + 1 \dots + 1}_{k = n} = n$$

⑥

```
for (i = n; i >= 1; i = i/2)
{
    stmt;
}
```

Assume $i < 1$

$$\therefore \frac{n}{2^k} < 1$$

$$\frac{n}{2^k} = 1$$

$$n = 2^k$$

$$k = \log_2 n$$

$$O(\log_2 n)$$

$$\frac{i}{n} \\ \frac{n}{2} \\ \frac{n}{2^2} \\ \frac{n}{2^3} \\ \dots \\ \frac{n}{2^k}$$

⑨

$p = 0$

for($i = 1; i < \underline{n}; i = i * 2$)

{

$p++;$ ————— $p = \log n$

}

for($j = 1; j < \vec{p}; j = j * 2$)

{

stmt;

}

————— $\log p$

$\uparrow$
 $O(\log \log n)$

for($i=0; i < n; i++$) — n
{

for($j=1; j < n; j=j*2$) — $n \times \log n$
{

stmt; — $n \times \log n$
}

}

$$\frac{2n \log n + n}{O(n \log n)}$$

Types of Time functions

$O(1)$ — Constant

$O(\log n)$ — Logarithmic

$O(n)$ — Linear

$O(n^2)$ — Quadratic

$O(n^3)$ — Cubic

$O(2^n)$ — Exponential

$O(3^n)$

$O(n^n)$

$$f(n) = 2 \rightarrow O(1)$$

$$f(n) = 5 \rightarrow O(1)$$

$$f(n) = 5000 \rightarrow O(1)$$

$$f(n) = 2n + 3 \rightarrow O(n)$$

$$f(n) = 500n + 700 \rightarrow O(n)$$

$$f(n) = \frac{n}{5000} + 6 \rightarrow O(n)$$

$$< \log n < \sqrt{n} < n < n \log n < n^2 < n^3 < \dots < 2^n < 3^n < \dots < n^n$$

$\log n$	n	n^2	2^n
0	1	1	2
$\log_2^2 = 1$	2	4	4
2	4	16	16
3	8	64	256
3.1	9	81	512

$$n^{100} < 2^n$$

$$n^k < 2^n$$

